

## SINGAPORE PHYSICS OLYMPIAD 2020

### Practical Test 19 NOVEMBER 2020

This experiment section of the Physics Olympiad consists of 2 sections. You are given **two hours** for the experiment. Write your answers/data in this question paper. ANSWER ALL QUESTIONS.

Name: \_\_\_\_\_

Junior College/School: \_\_\_\_\_

Name:

Junior College/School:

### **Section (A) Determine the mass of Cartesian Diver inside a bottle of water**

The following figure shows a sealed bottle of water in which there is a sealed plastic tube partially filled with coloured water. This plastic tube is known as a Cartesian Diver.



Figure: Cartesian Diver inside a bottle of water.

### **Section (A1) : How does the Cartesian diver work?**

**Section (A1)** Squeeze the sealed bottle of water with the Cartesian Diver in it and then un-squeeze it.

Describe your observations.

Explain your observations.

Name:

Junior College/School:

### **Section (A2) Determine the mass of the Cartesian Diver**

Design and carry out an experiment or a series of experiments to determine the mass of the Cartesian Diver inside the sealed bottle. Note that you are not allowed to remove the Cartesian Diver from the sealed bottle. You are not allowed to open the cap to the sealed bottle to put items into the bottle. And you are only allowed to use the items provided and as shown in the picture below. You can of course use your calculator.

**Hint: you may or may not need to make use of all the items provided. Some items are meant to distract you.**



Name:

Junior College/School:

**Section (A2)**

Describe what you plan to do here.

Name:

Junior College/School:

**Section (A2)** Carry out the experiment(s) you have designed and proceed to determine the mass of the Cartesian Diver.

The mass of the Cartesian Diver is determined to be

|  |       |  |   |
|--|-------|--|---|
|  | $\pm$ |  | g |
|--|-------|--|---|

Name:

Junior College/School:

**Section (B) Determine the mass of a ball inside a sealed tube**

The following figure shows a sealed plastic tube with a ball inside.



Design and carry out an experiment or a series of experiments to determine the mass of the ball. Note that you are not allowed to remove the ball from the sealed plastic tube. You are not allowed to open the cap to the sealed plastic tube to put items into the plastic tube. And you are only allowed to use the items provided (see page 3) and your calculator. **Hint: you may or may not need to make use of all the items provided. Some items are meant to distract you.**

**Section (B1)**

Describe what you plan to do here.

Name:

Junior College/School:

**Section (B2)** Carry out the experiment(s) you have designed and proceed to determine the mass of the ball.

The mass of the ball is determined to be

|  |       |  |   |
|--|-------|--|---|
|  | $\pm$ |  | g |
|--|-------|--|---|

Name:

Junior College/School: